Soviet system kept putting in the way of his project in Moscow. He was torn between loyalty to his native land and disgust with his native land, and between his feelings that life in America is barbaric because of the way we treat our poor and our lack of medical care for everyone and his feelings that life in Moscow is miserable because of the power of incompetent officials. He was torn between the freedom and wealth of America and fear of KGB retribution against family and friends and perhaps even himself if he "defected." In the end he said no, and recommended instead Ronald Drever of Glasgow University.

Others I consulted were also enthusiastic about Drever. Like Braginsky, he was highly creative, inventive, and tenacious—traits that would be essential for success of the project. The Caltech faculty and administration gathered all the information they could about Drever and other possible leaders, selected Drever, and invited him to join the Caltech faculty and initiate the project. Drever, like Braginsky, was torn, but in the end he said yes. We were off and running.

I had presumed, when proposing the project, that like Weber and Braginsky, Caltech would focus on building bar detectors. Fortunately (in retrospect) Drever insisted on a radically different direction. In Glasgow he had worked with bar detectors for five years, and he could see their limitations. Much more promising, he thought, were *interferometric* gravitational-wave detectors *(interferometers* for short—though they are radically different from the radio interferometers of Chapter 9).

Interferometers for gravitational-wave detection had first been conceived of in primitive form in 1962 by two Russian friends of Braginsky's, Mikhail Gertsenshtein and V.I. Pustovoit, and independently in 1964 by Joseph Weber. Unaware of these early ideas, Rainer Weiss devised a more mature variant of an interferometric detector in 1969, and then he and his MIT group went on to design and build one in the early 1970s, as did Robert Forward and colleagues at Hughes Research Laboratories in Malibu, California. Forward's detector was the first to operate successfully. By the late 1970s, these interferometric detectors had become a serious alternative to bars, and Drever had added his own clever twists to their design.

Figure 10.6 shows the basic idea behind an interferometric gravitational-wave detector. Three masses hang by wires from overhead supports at the